SECTION B

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
7.	(a)			**Indicative content** Trends • boiling temperatures increase down the groups • boiling temperature of HF is anomalous • Group 7 hydrides higher boiling temperatures than those of Group 4 Explanation • the stronger the intermolecular force the higher the boiling temperature • going down a group the number of electrons increases so van der Waals forces / induced dipole-induced dipole forces get stronger • HF has hydrogen bonds between molecules • hydrogen bonds are stronger than van der Waals forces so boiling temperature is much higher than expected • Group 7 hydrides are more polar so dipole-dipole forces present and these are stronger than induced dipole-induced dipole forces in Group 4 hydrides **5-6 marks** Trend down the groups and HF anomaly identified; trend from Group 4 to Group 7 identified; fully explained *The candidate constructs a relevant, coherent and logically structured method including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.* **3-4 marks** Trend down the groups and HF anomaly identified; attempt at explanation with reference to more than one type of intermolecular force *The candidate constructs a coherent account including most of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary are generally sound.*	3	2	1	6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-2 marks Trend/anomaly identified; reference to intermolecular forces *The candidate attempts to link at least two relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.* **0 marks** *The candidate does not make any attempt or give an answer worthy of credit.*						
	(b)	(i)		$CH_4 + 4Cu_2O \rightarrow 8Cu + CO_2 + 2H_2O$ accept multiples		1		1		
		(ii)		award (1) for any of following • oxidation number of carbon changes from –4 to +4 and copper changes from +1 to 0 • oxidation number of carbon increases and oxidation number of copper decreases • carbon loses electrons and copper gains electrons • carbon is oxidised and copper is reduced do not accept hydrogen is oxidised	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			Li : Al : H $\frac{38.7}{6.94} : \frac{50.1}{27.0} : \frac{11.2}{1.01}$ 5.58 : 1.86 : 11.1 (1) ⇨ 3 : 1 : 6 ⇨ empirical formula Li_3AlH_6 (1) award (1) for final answer Li_3Al where amount of hydrogen present has been overlooked		2		2	1	
				Question 7 total	**4**	**5**	**1**	**10**	**1**	**0**